

Potential Difference and Resistance *

Keywords: Voltage, volt, resistance, ohms, ohmic conductor

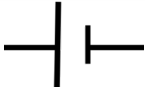
DO NOW: Complete these quiz questions about last lesson.

TIME FOR A

QUIZ!



Quiz:

1. What is this symbol? 
2. What is this symbol? 
3. What is the symbol for an ammeter?
4. What is the symbol for voltmeter?
5. What is this symbol?  a) LED b) resistor c) fuse d) Diode
6. Current is a flow of c_____
7. What is flowing are particles called e_____
8. Charge flow = c_____ x t_____
9. The unit for charge flow is a) amps b) volts c) coulombs d) joules
10. The unit for current is
a) amps b) volts c) coulombs d) joules

Potential Difference and Resistance *

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STARTER: Josie wants to test the voltage and the current with one bulb in the circuit. Draw the circuit she will have to construct.

EXT: Match up at least 2 circuit symbols to what the component is and the function.

circuit symbol

component

function

A



switch

breaks or makes
a circuit

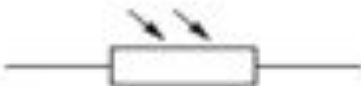
B



cell

resistance
varies with
temperature

C



thermistor

transforms
chemical energy
into electrical
energy

D

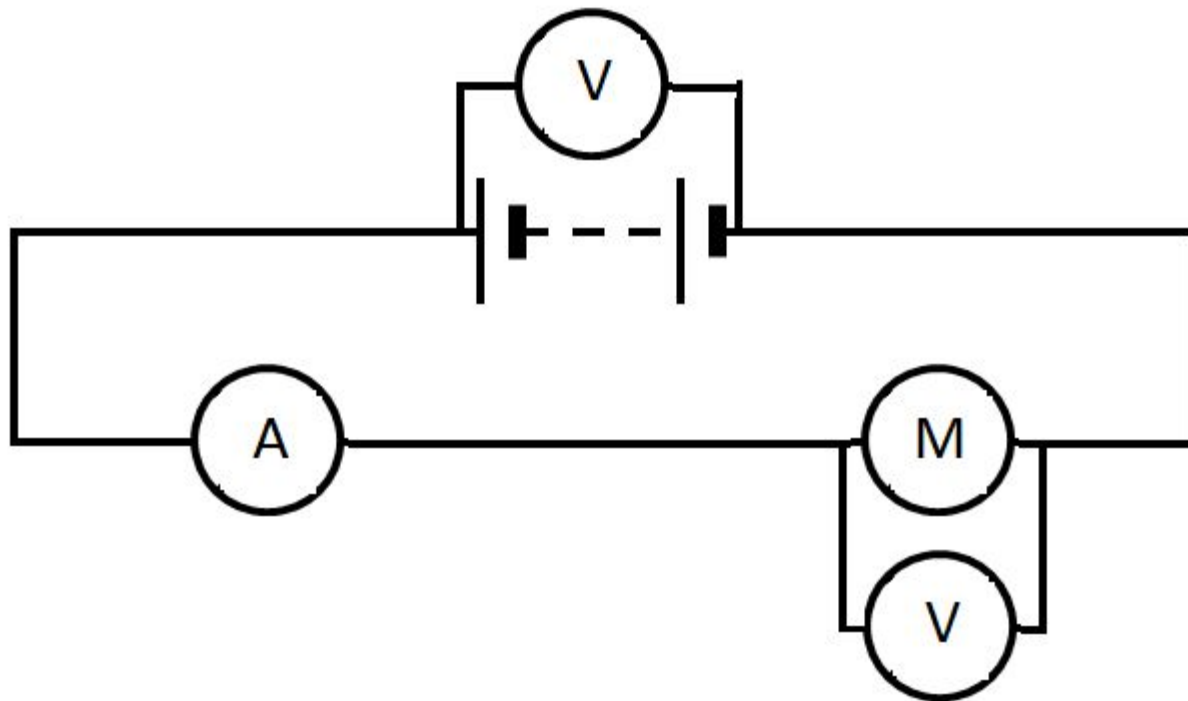
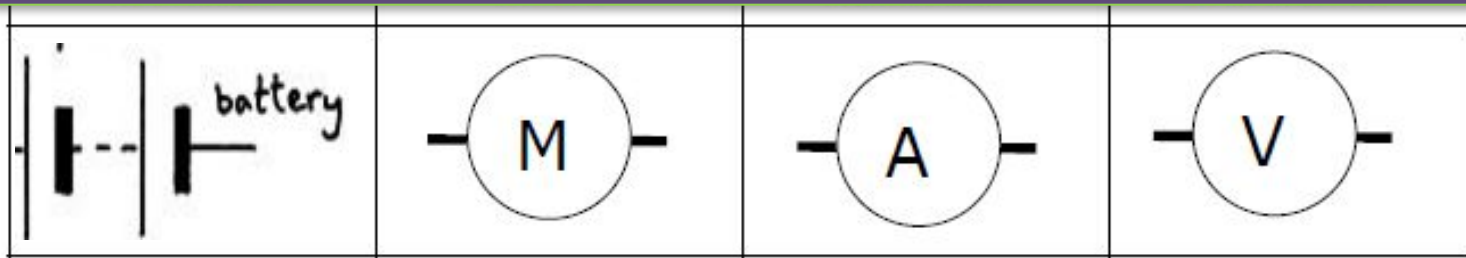


LDR

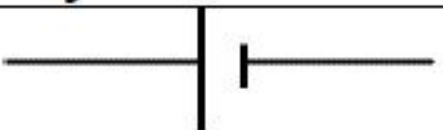
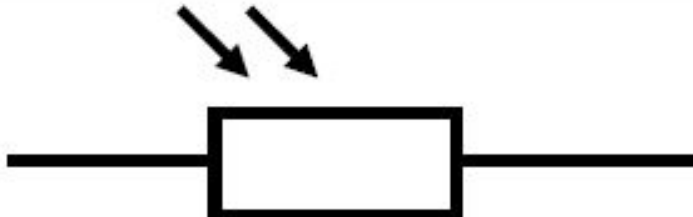
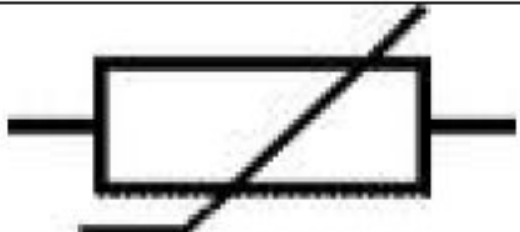
resistance
varies with light
intensity

1. Murtaza wants to test the voltage and the current with the motor in a circuit. Draw the circuit he will have to construct

2.



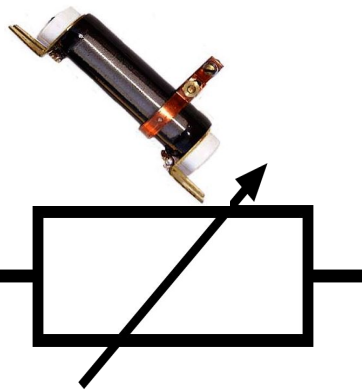
Review

		switch	breaks or makes a circuit
		cell	resistance varies with temperature
		thermistor	transforms chemical energy into electrical energy
		LDR	resistance varies light intensity

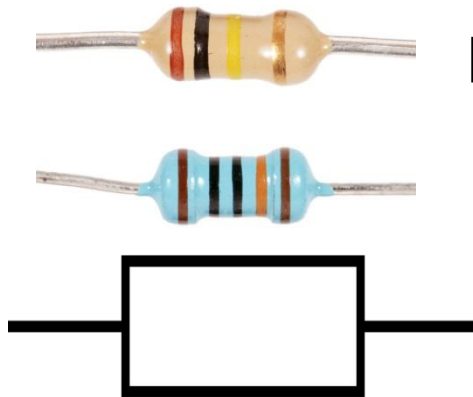
LOb: Understand the terms potential difference and resistance.

Keywords: Voltage, volt, resistance, ohms, ohmic conductor

Variable Resistor



Resistor



Success Criteria:

- Describe what is meant by potential difference
- Describe resistance and what its unit is
- Explain what happens when you reverse the potential difference across a resistor

In KS3 you learn that potential difference is measured in volts ✓

You also learn that a battery or cell provides the **push** needed to make the charge move around the circuit. This push is called potential difference. 😐

WATCH AND

LEARN

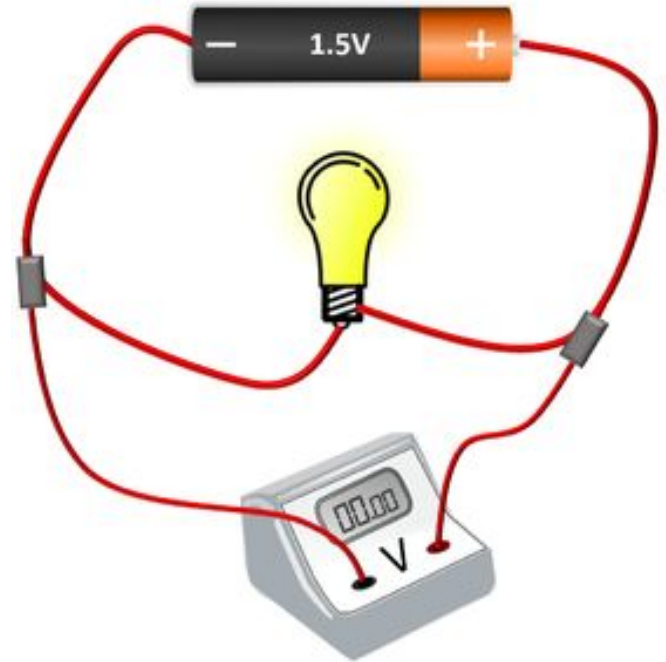
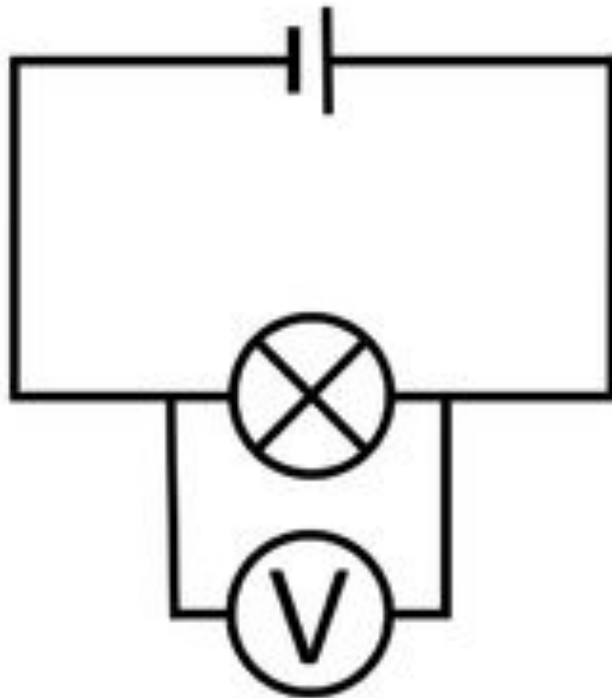
AFTERALL KNOWLEDGE IS

POWER!



Potential difference is measured in **volts (V)** using an **voltmeter**.

The voltmeter is attached parallel with the component being tested.

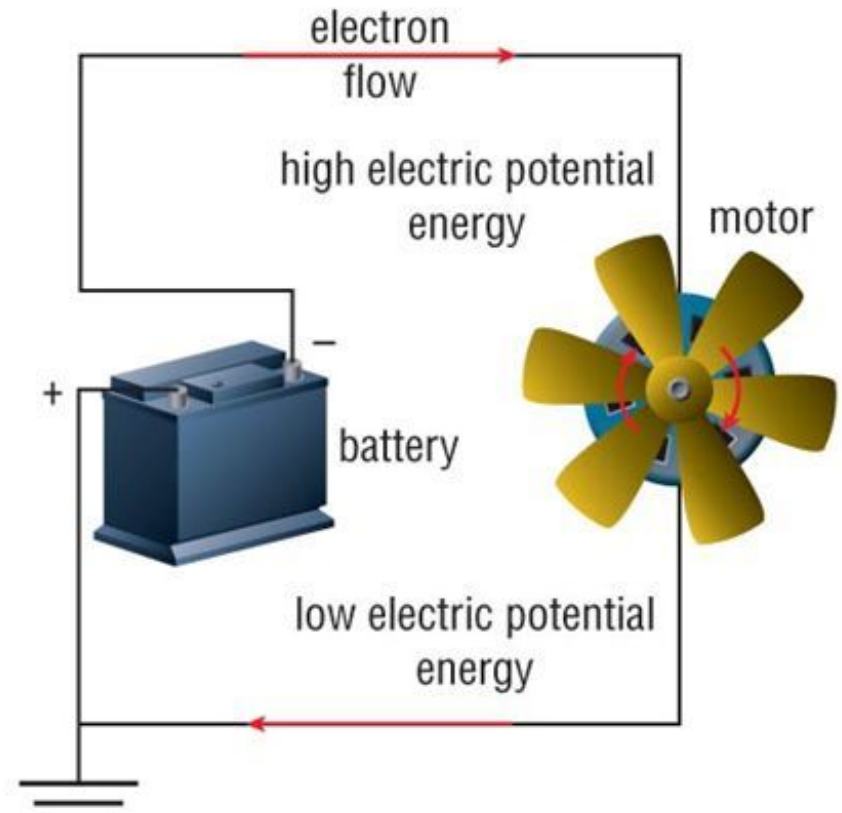
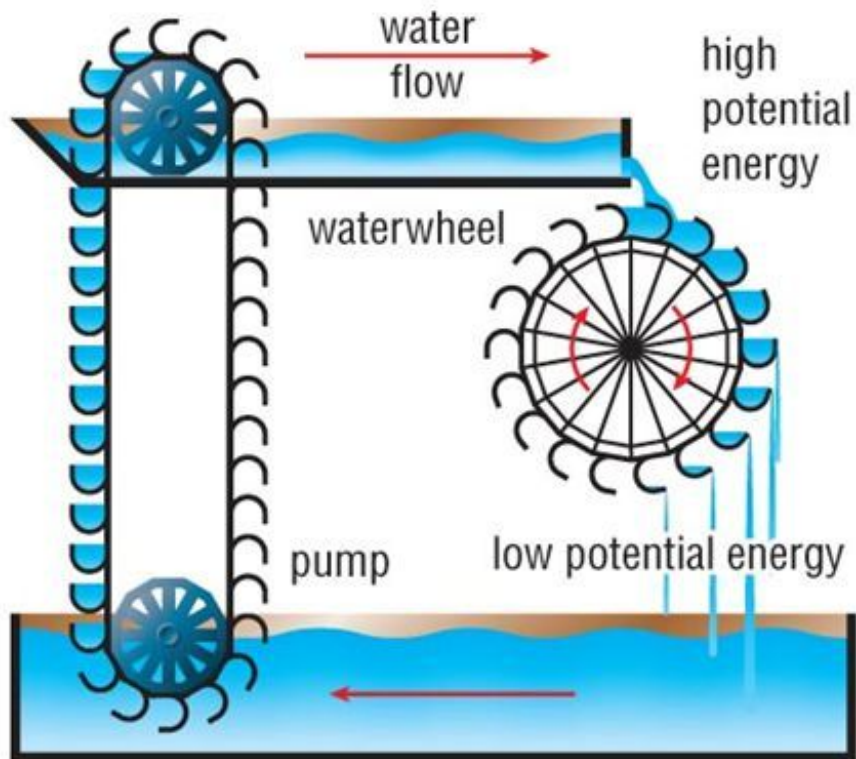


Potential difference (Volts, V) is the **work done** by the **component per coulomb of charge** passing through it.

What represents the current flow in this analogy?

What is the component?

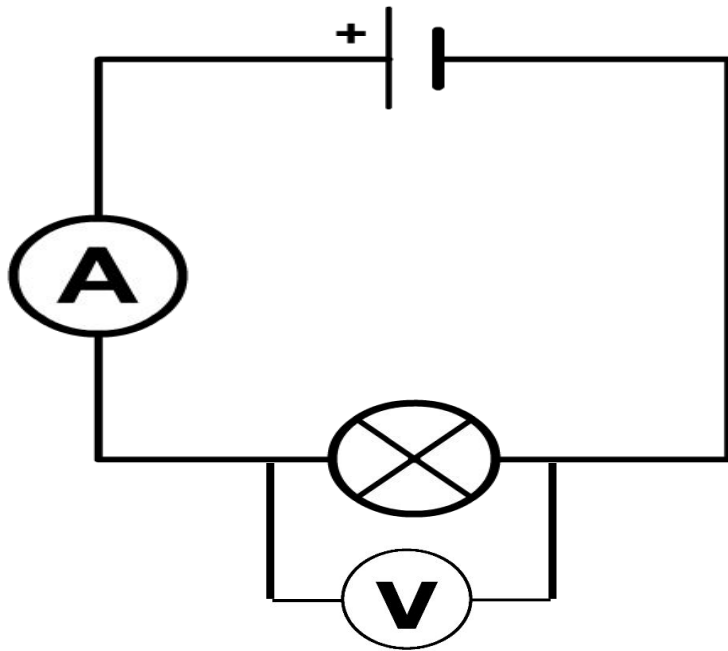
What represents the battery?



Potential difference equation.



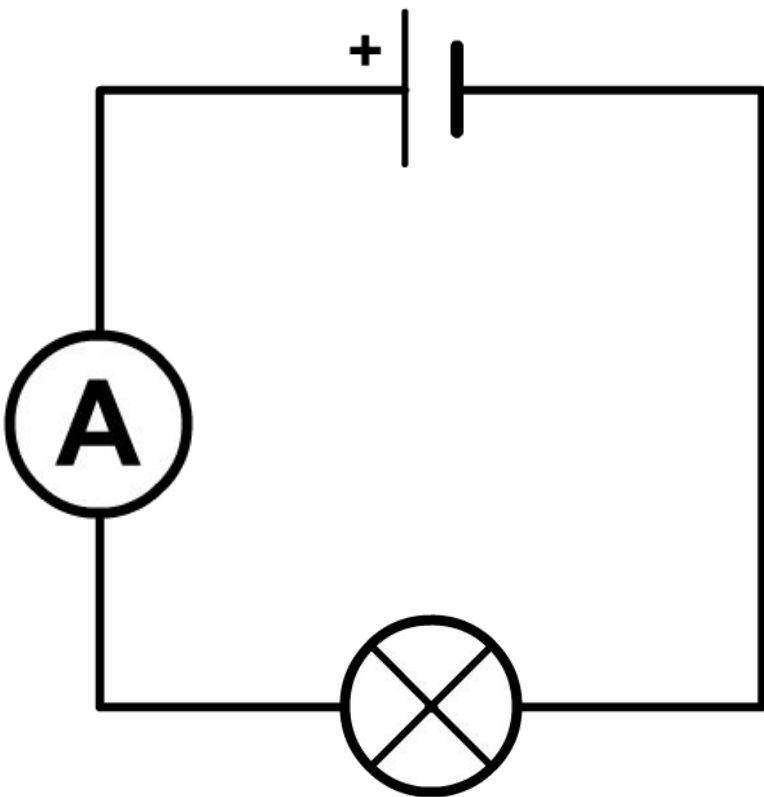
$$\text{Potential difference across Component, } V \text{ (Volts, V)} = \frac{\text{Energy transferred, } E \text{ (Joules, J)}}{\text{Charge, } Q \text{ (coulombs, C)}}$$



Worked Example:

The energy transferred to a bulb is 320 J when 64 C of charge is passed through it. Calculate the potential difference

$$\text{Potential difference across Component, } V \text{ (Volts, V)} = \frac{\text{Energy transferred, } E \text{ (Joules, J)}}{\text{Charge, } Q \text{ (coulombs, C)}}$$



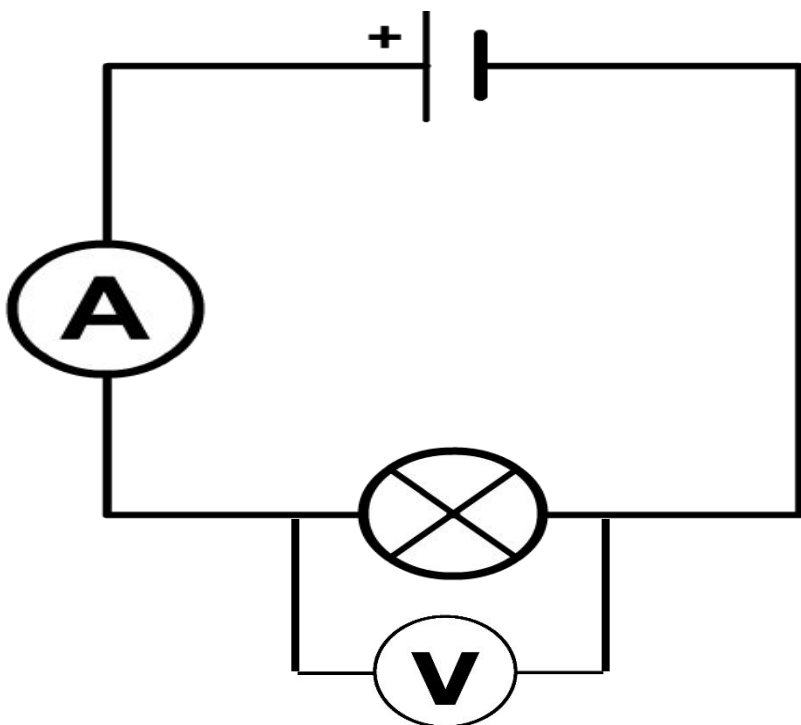
$$V = \frac{E}{C}$$

↓

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↓

$$V = \frac{E}{C}$$

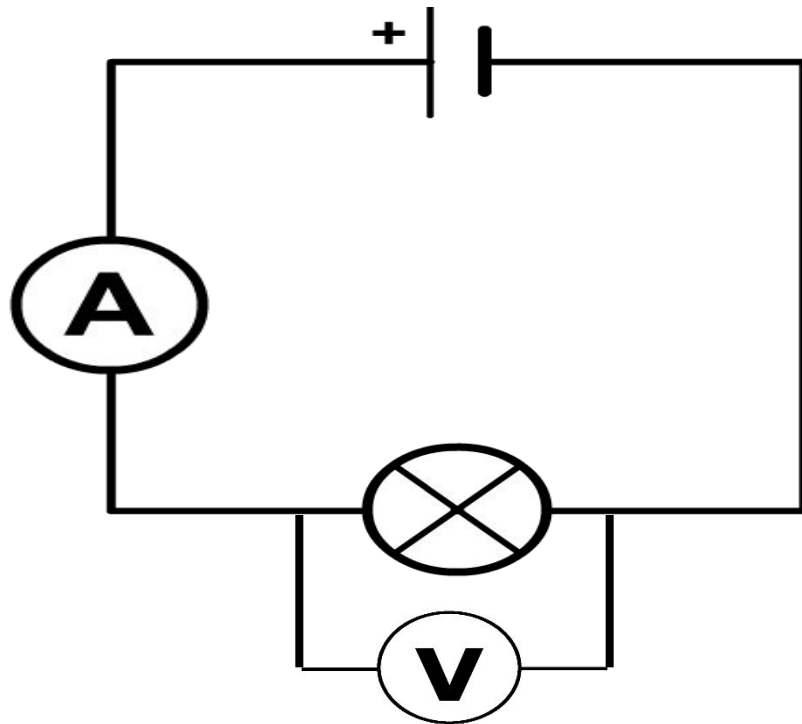


The voltmeter measures the _____

The analogy of the _____ explains potential difference.

$$\text{Potential difference across } \underline{\hspace{2cm}}, V \text{ (Volts, V)} = \frac{\underline{\hspace{2cm}}, E \text{ (Joules, J)}}{\text{Charge, } Q \text{ (} \underline{\hspace{2cm}} \text{, C)}}$$

Potential difference explains the amount of work _____ done _____ by each coulomb of charge.



The voltmeter measures the _____
p.d. in volts

The analogy of the **water wheel** explains potential difference.

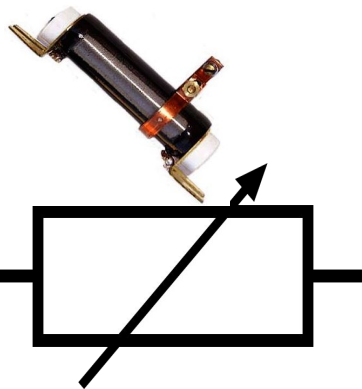
$$\text{Potential difference across component } V \text{ (Volts, V)} = \frac{\text{Energy transfer } E \text{ (Joules, J)}}{\text{Charge, } Q \text{ (coulombs C)}}$$

Potential difference explains the amount of **work done** _____ by each coulomb of charge.

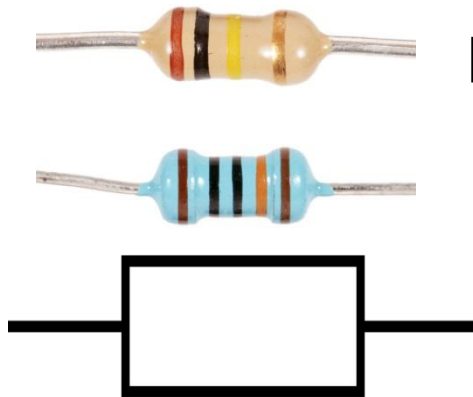
LOb: Understand the terms potential difference and resistance.

Keywords: Voltage, volt, resistance, ohms, ohmic conductor

Variable Resistor



Resistor



Success Criteria:

- ✓ Describe what is meant by potential difference
- Describe resistance and what its unit is
- Explain what happens when you reverse the potential difference across a resistor

Resistance

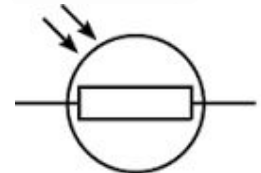
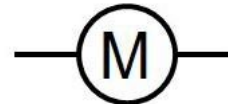
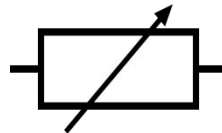
An **electrical** quantity that measures how the device or material reduces the **electric** current flow through it.

Measured in Ω
Ohms

COMPONENTS

CAUSE

RESISTANCE



You have thousands of runners.



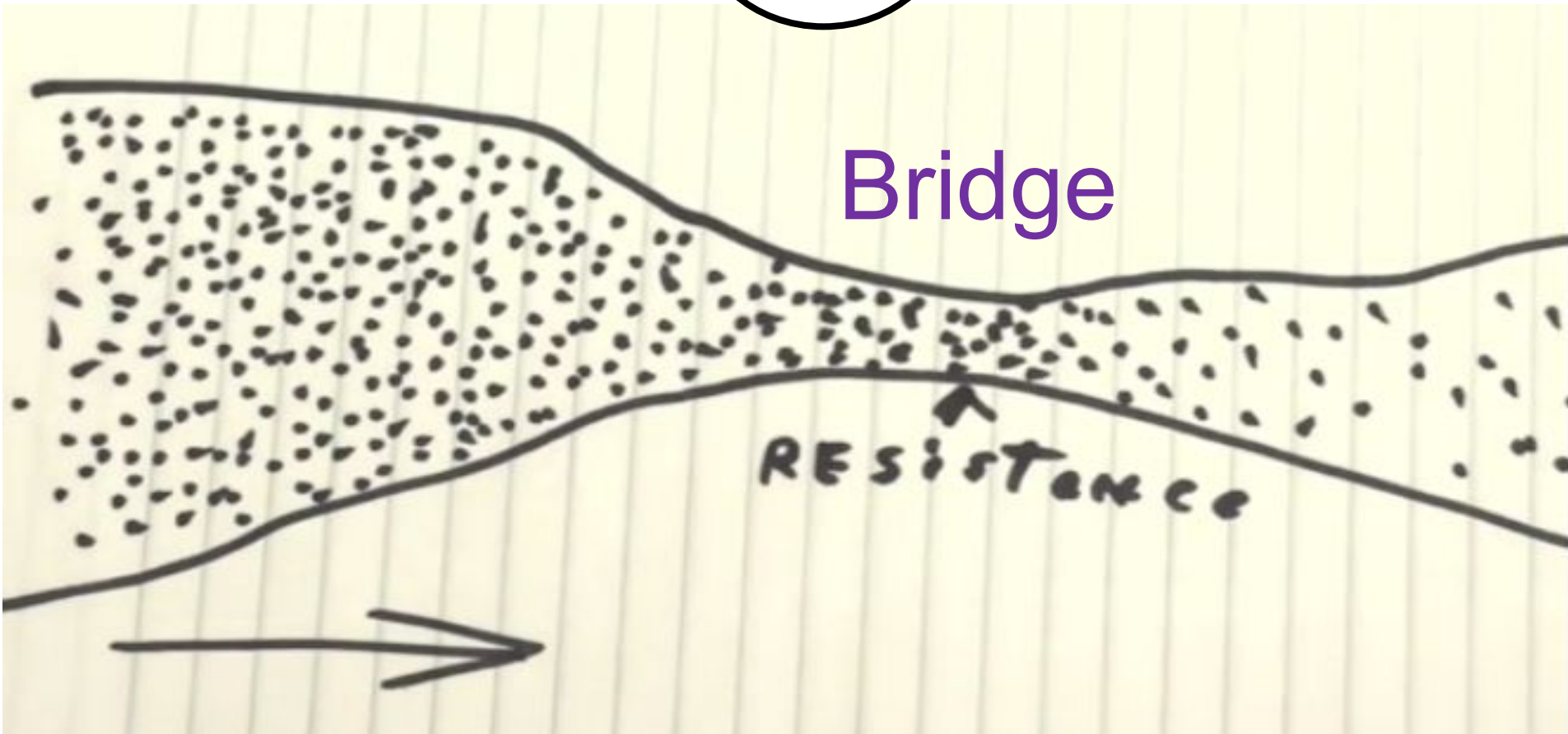
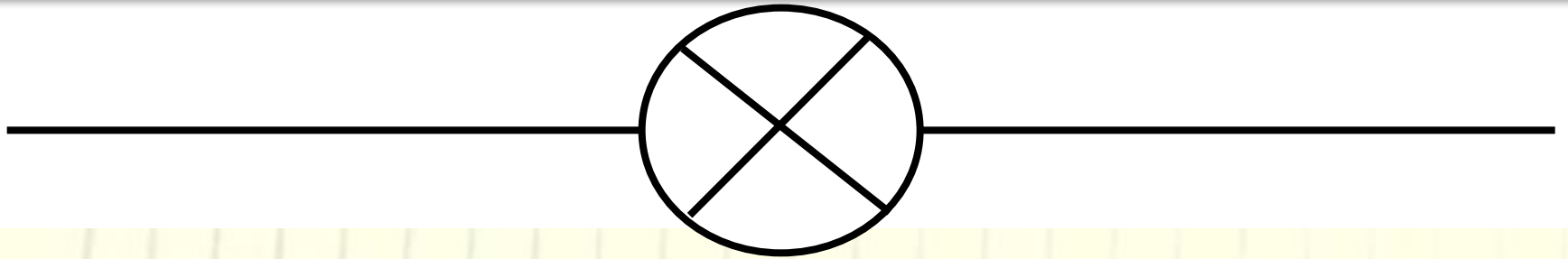
Up ahead is this bridge.



What will happen?

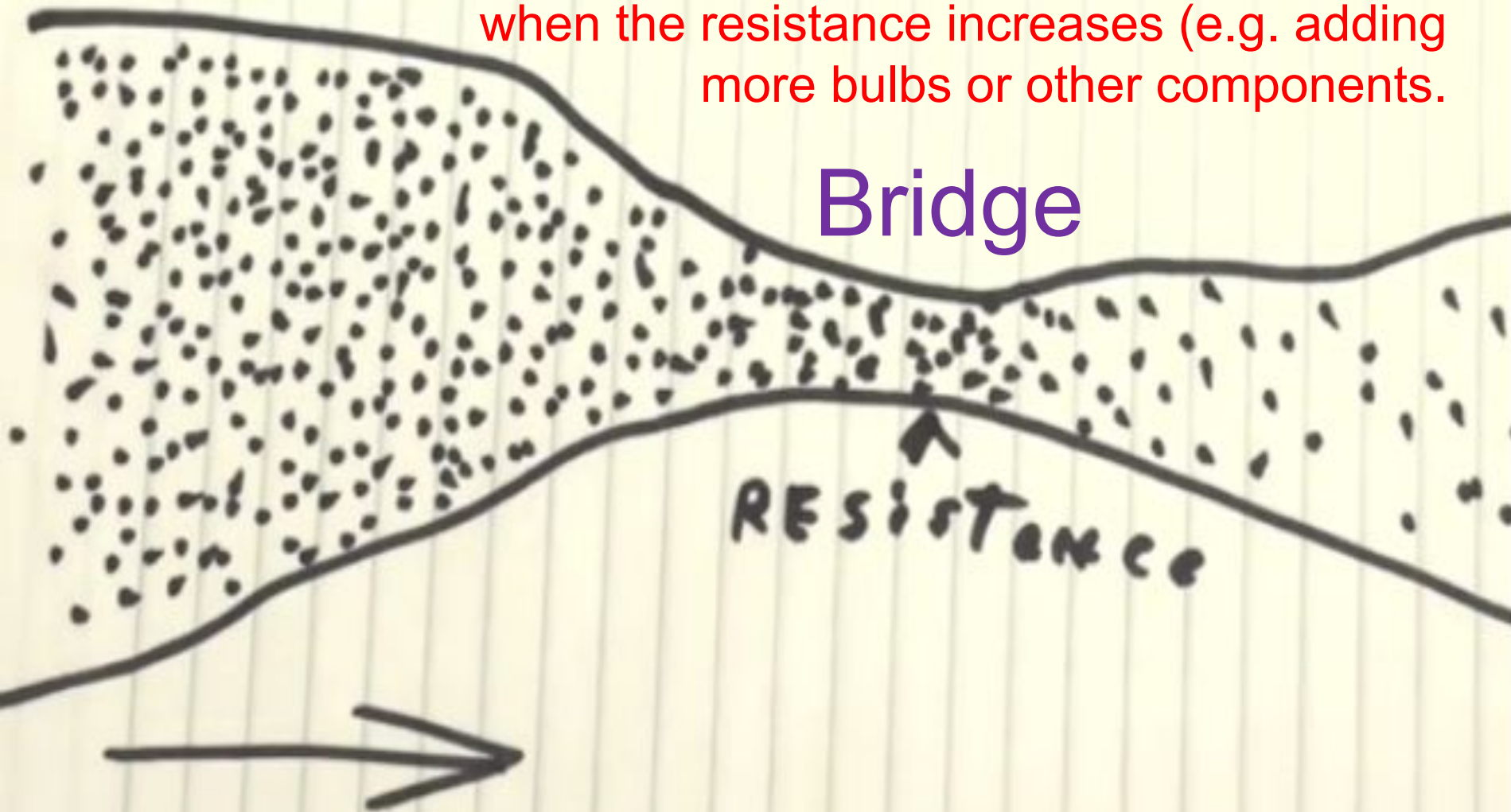


Electrons approach at speed they won't/can't slow down.
They will bump into each other and scrape on the edges of
the bridge to get through.

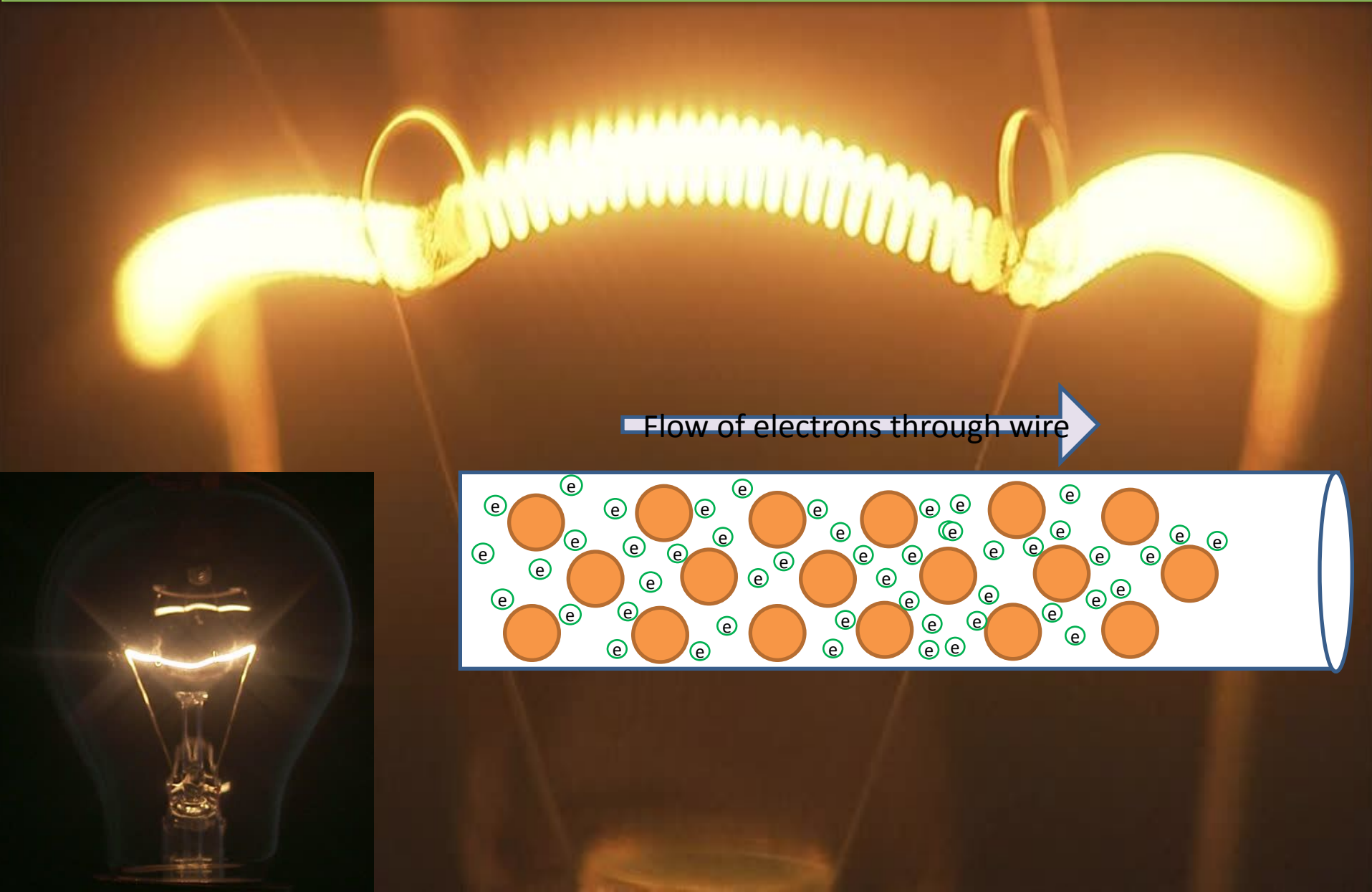


Electrons get through at the same speed but there aren't as many getting through the bridge

That is why the current goes down when the resistance increases (e.g. adding more bulbs or other components).



Electrons passing through a torch bulb have to push their way through lots of vibrating atoms in the metal filament.



Resistance causes friction

Friction is caused by the electrons colliding with the atoms in the wire. High friction causes **heat** and **light** to be produced.



1. What is resistance measured in?

2. How can we increase the resistance in a circuit?

3. Resistance stops the flow of what?

4. Describe resistance using the marathon runner analogy?

Runners represent the flow of _____

The road represents the?

The bridge represents the?

5. Draw a diagram of electrons passing through atoms in a wire



1. What is resistance measured in? **Ohms**

2. How can we increase the resistance in a circuit? **Adding components**

3. Resistance stops the flow of what? **Current (Electrons)**

4. Describe resistance using the marathon runner analogy?

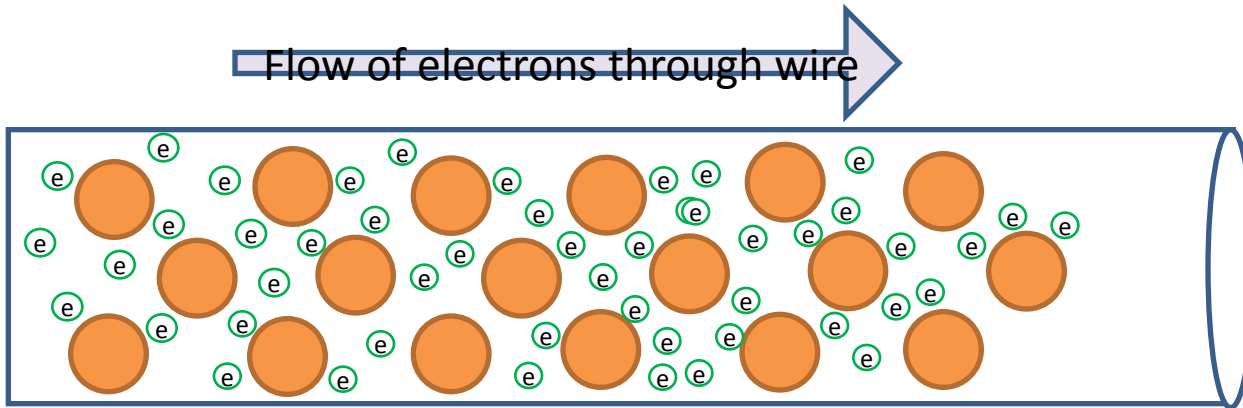
Runner represent the flow of current

The road represents the wire

The bridge represents the component



What factors might affect the resistance?



- ☐ Length of wire
- ☐ Thickness of wire
- ☐ Type of wire
- ☐ Temperature of wire

Resistance of an electric component is defined as...



Higher Tier

- ☐ **Recall**
- ☐ **Use**
- ☐ **Rearrange**

Resistance (R) = Potential Difference (V) ÷ Current (I)

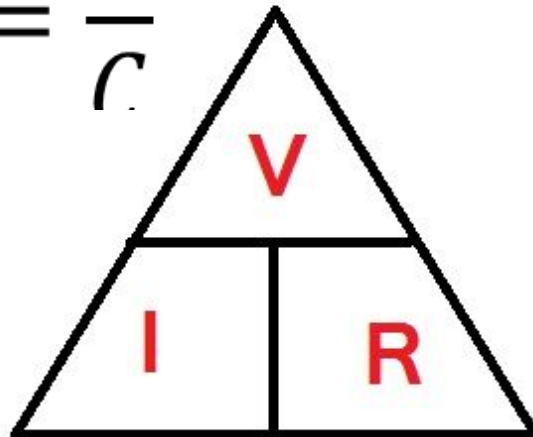
$$V = \frac{E}{I}$$

$$V = \frac{E}{I}$$

$$V = \frac{E}{I}$$



$$V = \frac{E}{I}$$



Foundation Tier

- ☐ **Use**
- ☐ **Rearrange**

Resistance of an electric component is defined as...

equation

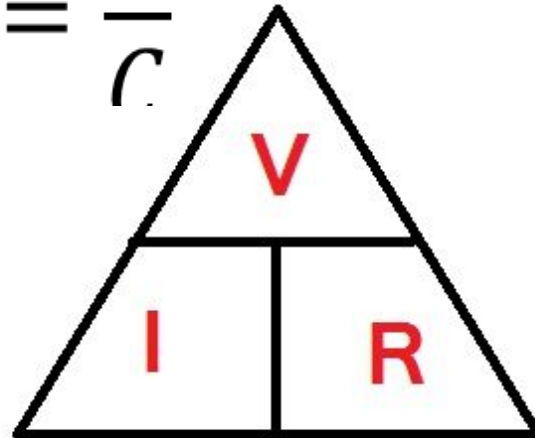
Higher Tier

- ☐ **Recall**
- ☐ **Use**
- ☐ **Rearrange**

$$V = \frac{E}{I} = \frac{\text{Potential difference, } V \text{ (volts, V)}}{\text{Current, } I \text{ (amperes, A)}}$$



$$V = \frac{E}{I}$$



Foundation Tier

- ☐ **Aided recall**
- ☐ **Use**
- ☐ **Rearrange**

Resistance (R) = Potential Difference (V) ÷ Current (I)

$$V = \frac{E}{I}$$

$$V = \frac{E}{I}$$

$$V = \frac{E}{I}$$

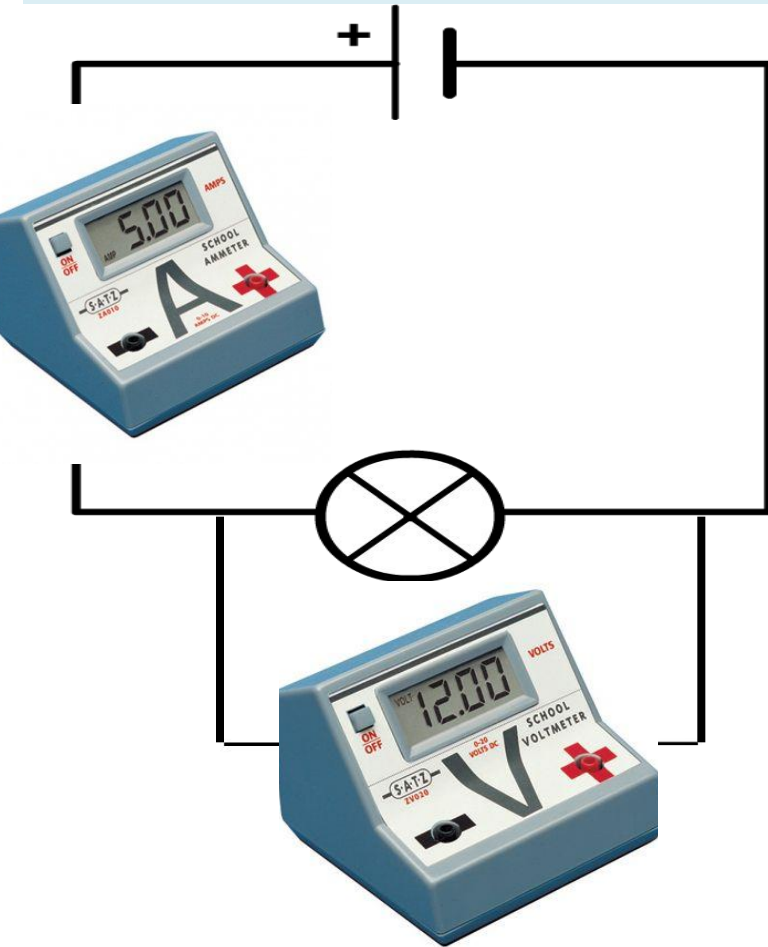
Rearrange the equation to find...

a) Current $V = \frac{E}{I}$

b) Potential difference $V = \frac{E}{I}$

Activity: Answer these calculation questions

2.1. A person set up this circuit and collected these results. Calculate the resistance of the circuit

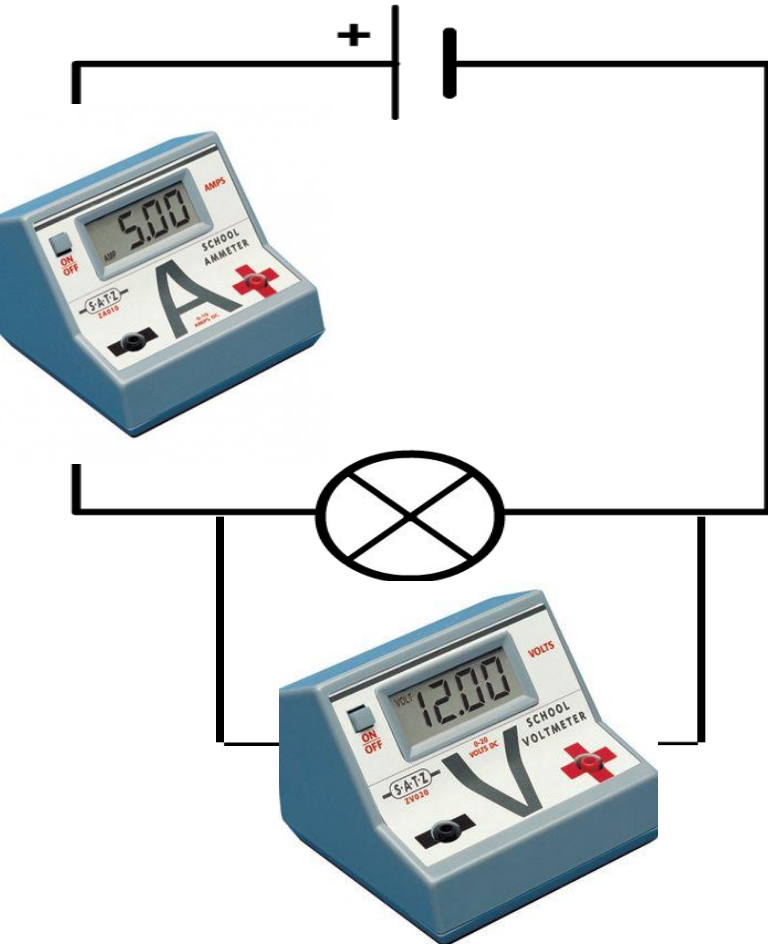


Answer _____ Unit _____

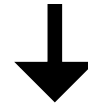
2.2. Calculate the current if the resistance of the circuit is 3.4 ohms and the potential difference is 27.5. Show your answer to 2 decimal places.

Review

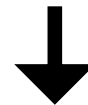
2.1. A person set up this circuit and collected these results. Calculate the resistance of the circuit



$$V = \frac{E}{C}$$



$$V = \frac{E}{C}$$



$$V = \frac{E}{C}$$

Review

2.2. Calculate the current if the resistance of the circuit is 3.4 ohms and the Potential difference is 27.5. Show your answer to 2 decimal places.

$$V = \frac{E}{C} \rightarrow V = \frac{E}{C} \rightarrow V = \frac{E}{C}$$
$$\downarrow$$
$$V = \frac{E}{C}$$

Ω

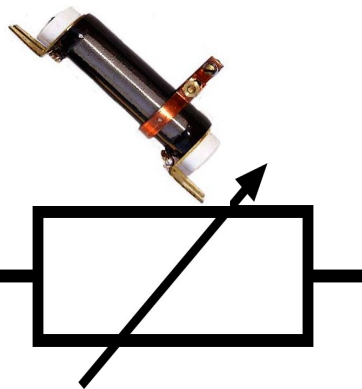
Activity: Answer the resistance questions on your sheet.



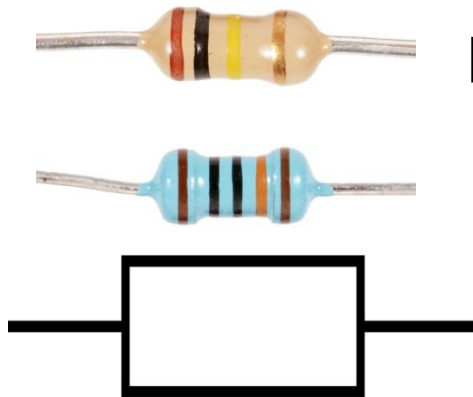
LOb: Understand the terms potential difference and resistance.

Keywords: Voltage, volt, resistance, ohms, ohmic conductor

Variable Resistor



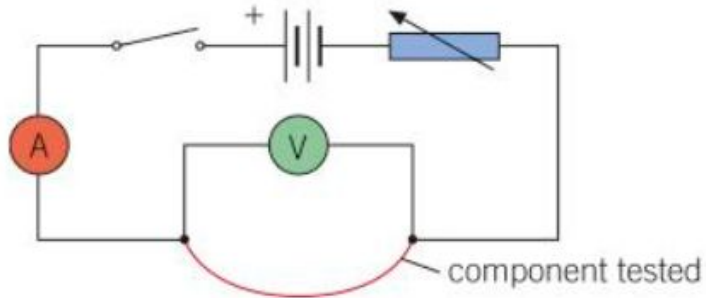
Resistor



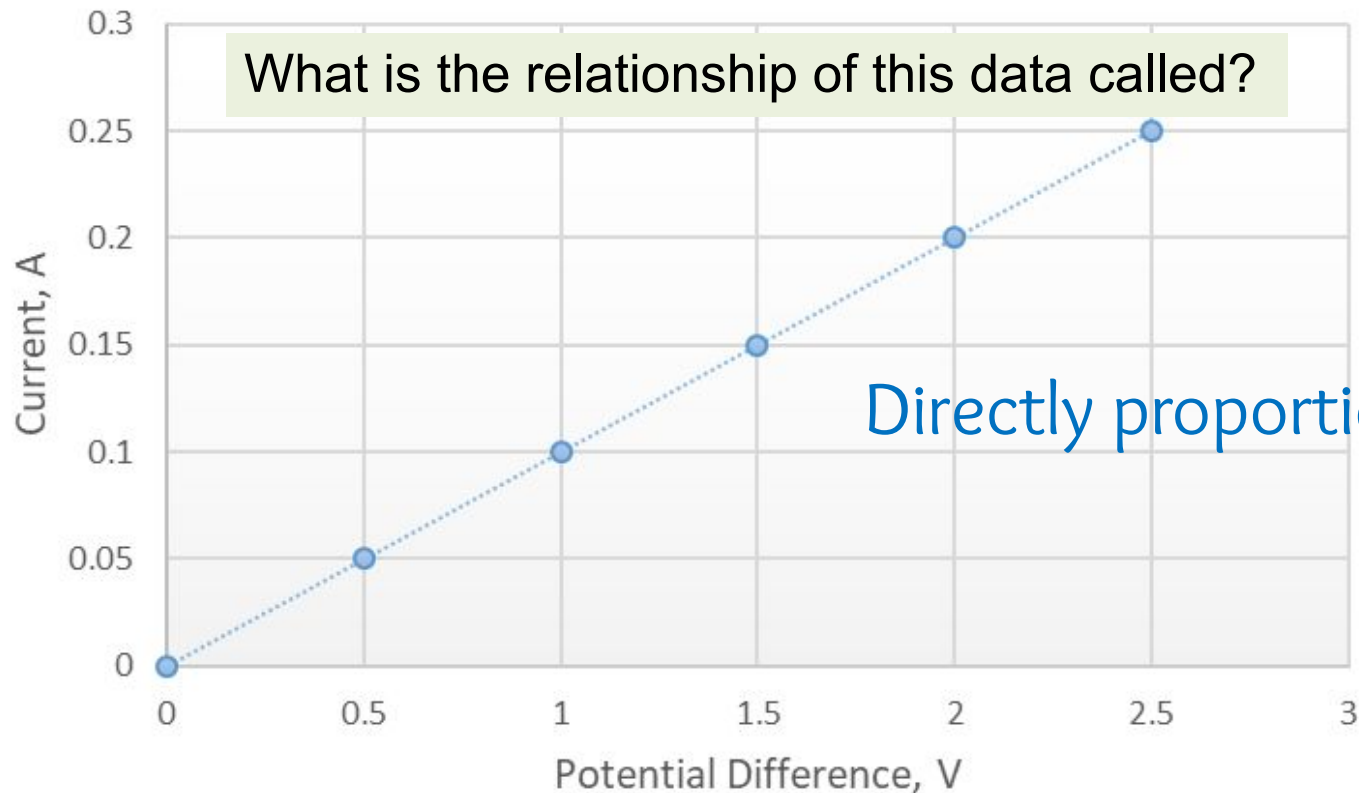
Success Criteria:

- ✓ Describe what is meant by potential difference
- ✓ Describe resistance and what its unit is
- Explain what happens when you reverse the potential difference across a resistor

A **wire was tested** in a circuit to see what happens to the current and potential difference. The graph of results is below.



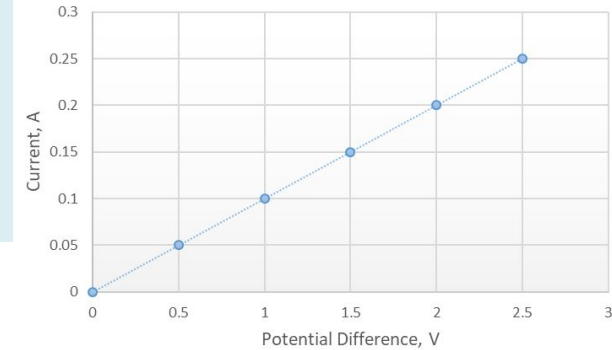
In a wire, when potential difference in a circuit increases, the current increases. — *as long as the temperature remains constant.*





Georg Ohm discovered that the current through a resistor is directly proportional to the potential difference across the resistor. This means the resistance doesn't change provided the temperature remains constant:

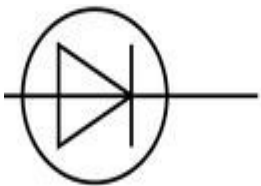
$$V = \frac{E}{C}$$



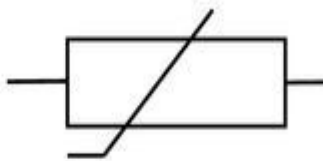
Ohm's

S

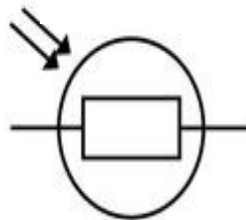
The current through a component is proportional to the p.d. across it (provided the temperature remains constant).



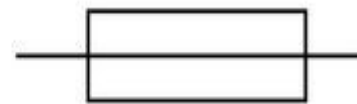
Diode



Thermistor



LDR



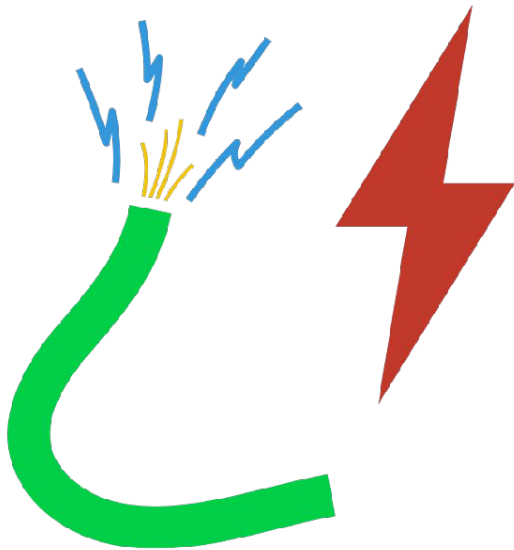
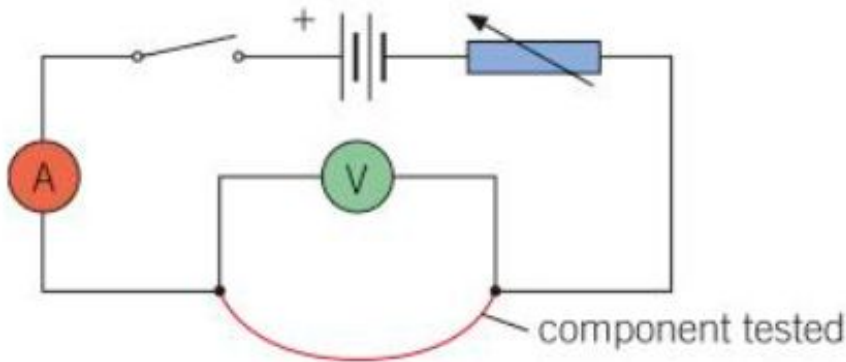
Fuse



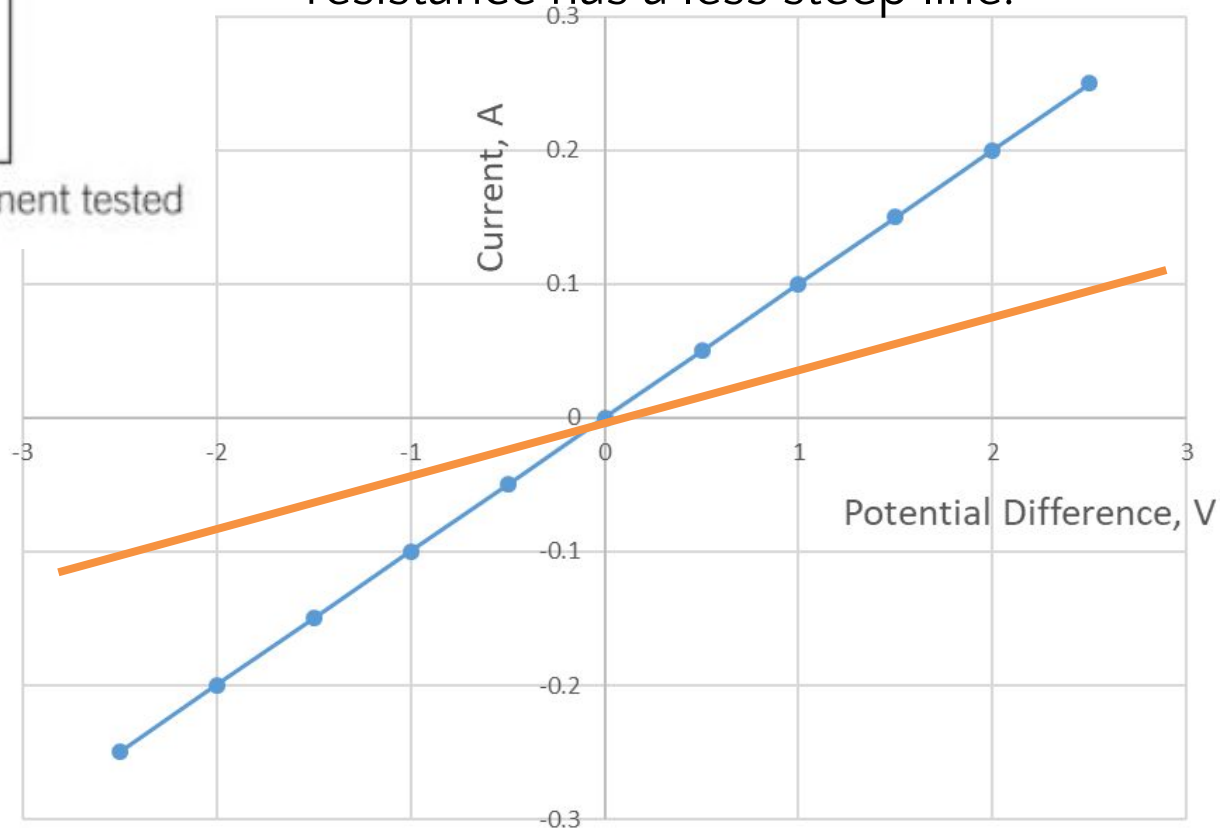
Resistor

Wires are called **ohmic conductors** because they obey ohms law.

If we reverse the potential difference (and therefore the current) the relationship is the same. The relationship is still directly proportional



A wire component with a greater resistance has a less steep line.



Activity: Answer these review questions

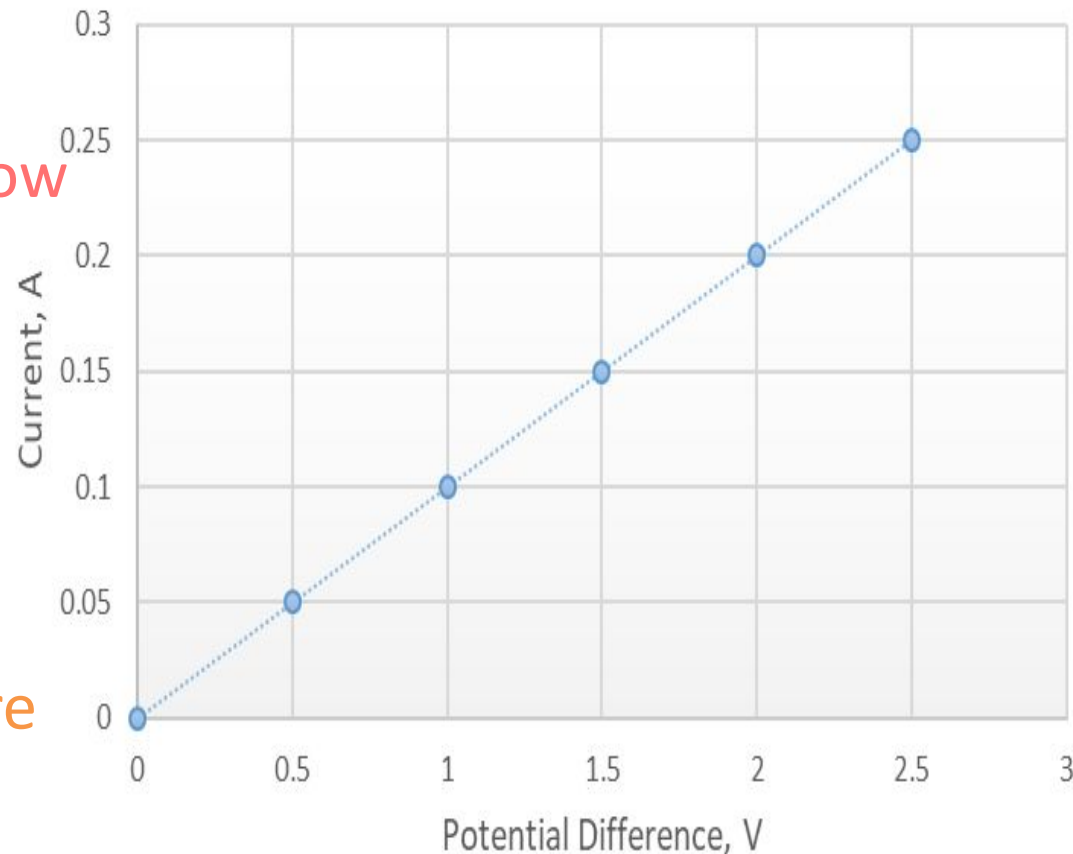
Question 1. What is ohms law?

Missing word The graph shows potential difference is _____ to the current.

Calculate Calculate the resistance of any point on the graph.

Sketch Draw this graph. Then add another line to show how it would look if the resistor had less resistance

Stretch Explain why the resistance increases when the temperature of the wire increases.



Review

Question

The current through a resistor is directly proportional to the potential difference across the resistor.

Missing word

The graph shows potential difference is directly proportional to the current.

Calculate

$$V = \frac{E}{C}$$

$$V = \frac{E}{C}$$

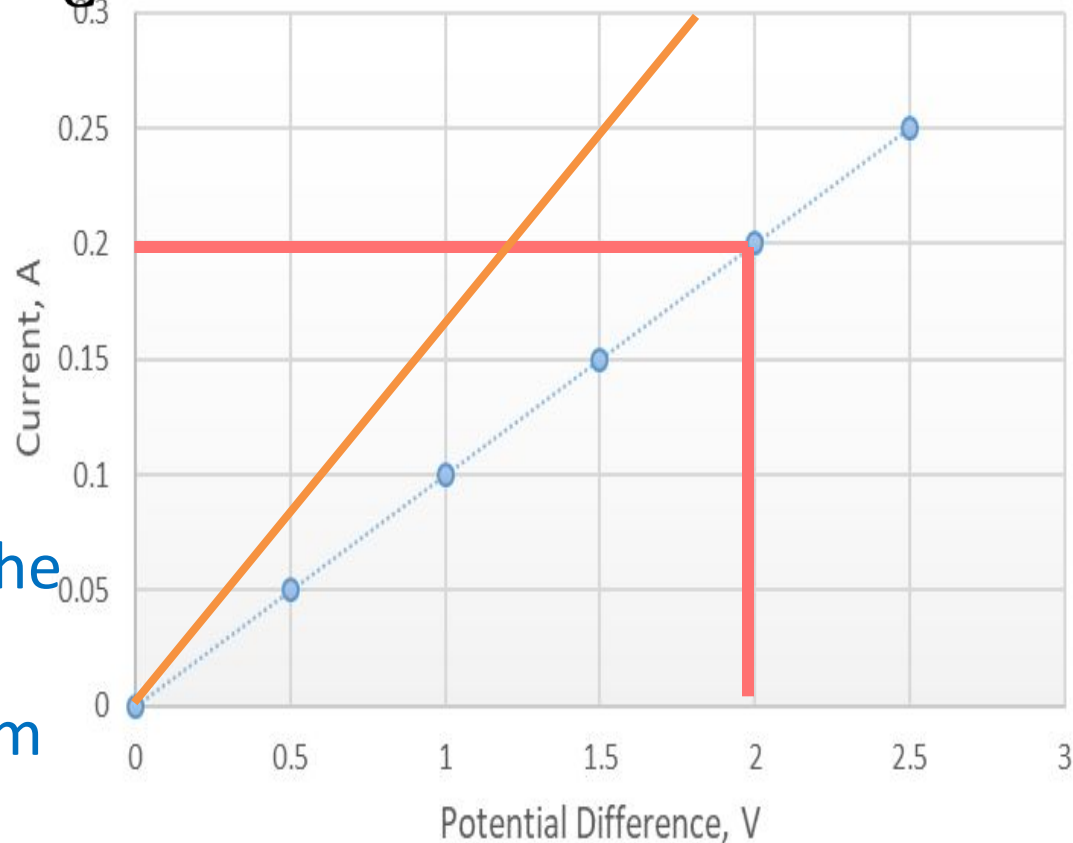
$$V = \frac{E}{C}$$

Sketch

The line is steeper

Stretch

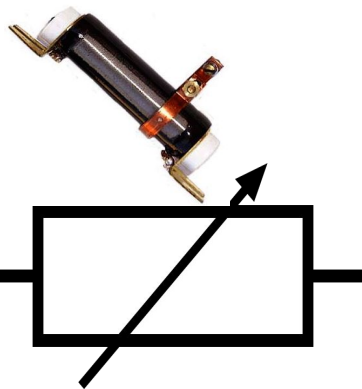
Resistance increase when the temperature of the wire increases because atoms in the wire vibrate more vigorously which prevents electrons from passing through as easily.



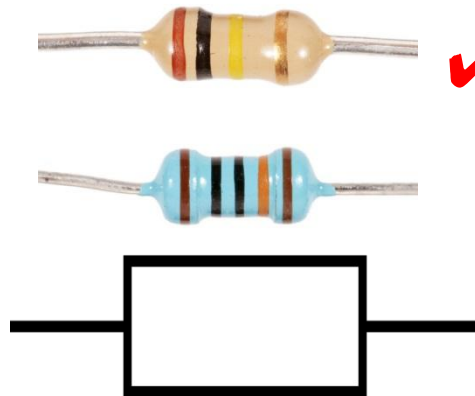
LOb: Understand the terms potential difference and resistance.

Keywords: Voltage, volt, resistance, ohms, ohmic conductor

Variable Resistor



Resistor



Success Criteria:

- ✓ Describe what is meant by potential difference
- ✓ Describe resistance and what its unit is
- ✓ Explain what happens when you reverse the potential difference across a resistor

keyfacts



Two equations:

Potential difference, V = energy $\div$ charge

$$V = \frac{E}{C}$$

Resistance, Ω = Potential difference $\div$ current

$$V = \frac{E}{C}$$

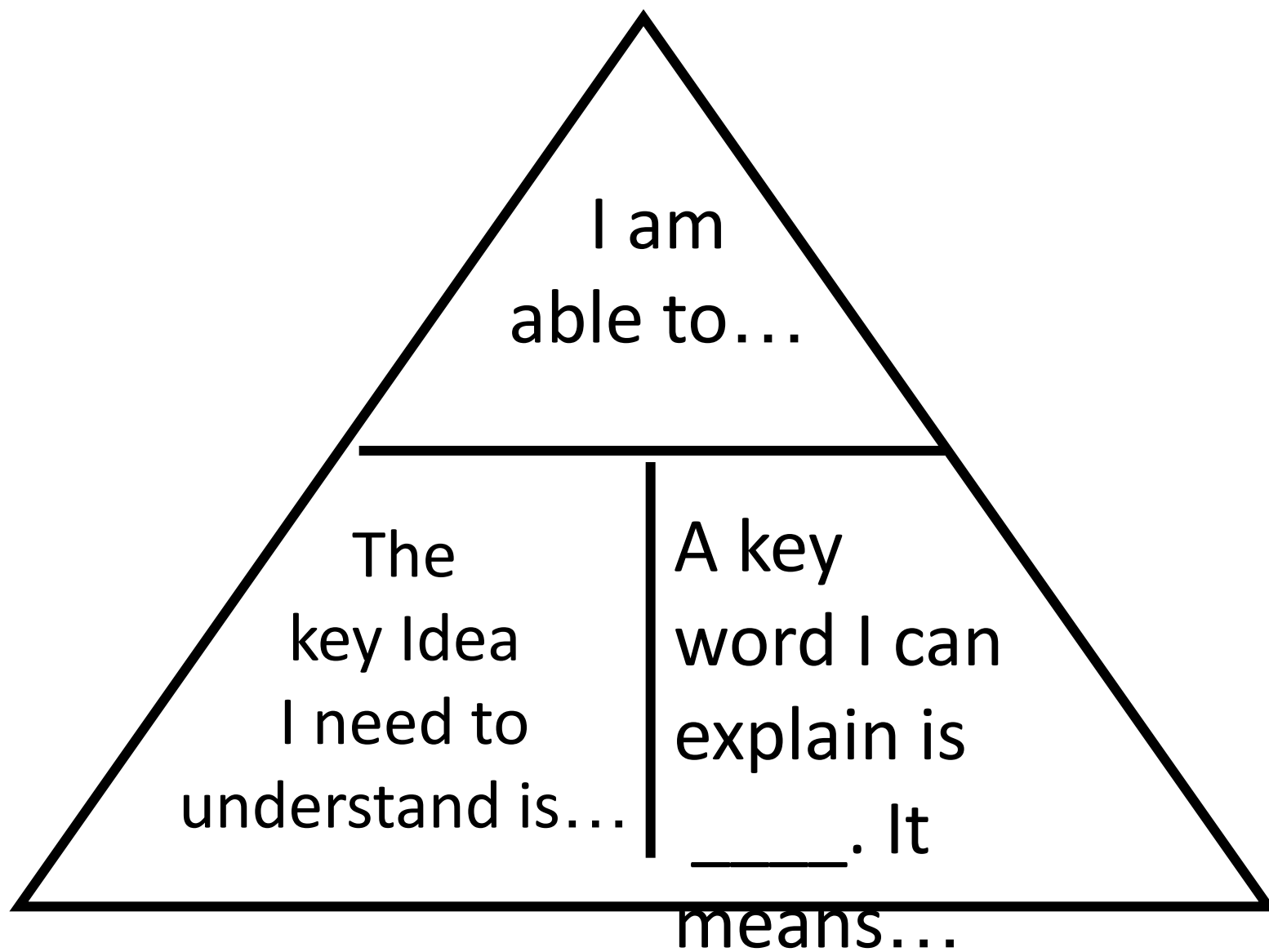


The current through a resistor is **directly proportional** to the potential difference across the resistor if the temperature remains constant. This is called **ohm's law**.



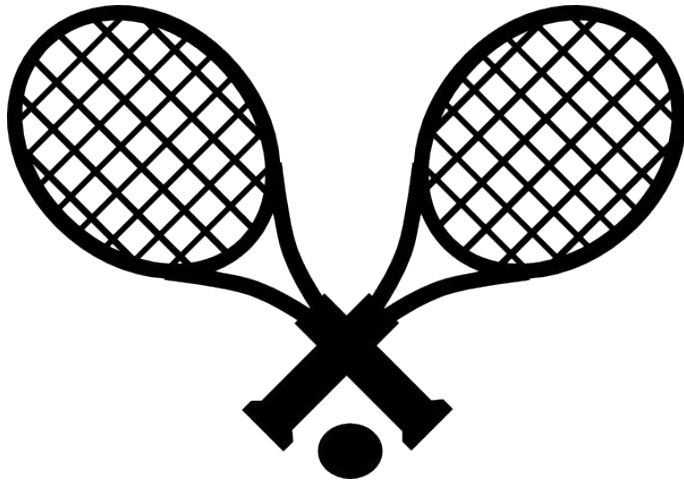
Reversing the potential difference reverses the current.

Complete the all questions in this learning triangle.



Plenary: Key Word Tennis

Make a list of as many keywords as possible from the lessons we have studied



Resistance (required practical) *

Keywords: Resistance, ohms, conductor, insulator

Starter: Using these bits of equipment below describe how you could investigate the resistance of a circuit.

FACT: As you change the length of wire the resistance changes.

VARIABLES

Independent - what we change

Dependent - What we measure

Control - what we keep the same



LOb: Understand and investigate the resistance in a circuit

Keywords: resistance, Ohm's law, current, potential difference (voltage), electrons

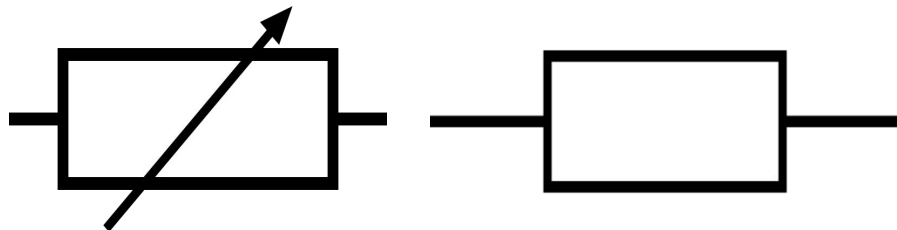
Learning outcomes:

- Recall what resistance is and its unit
- Investigate the resistance in a circuit making note of variables and the safety

Variable Resistor



Resistor



Did you know... resistors come in different forms but all have a basic job. The variable resistor can change its resistance

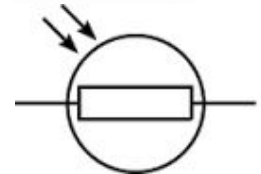
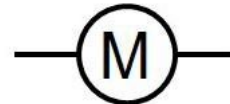
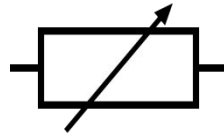
Resistance

*measures how a device or material reduces the current flow through it.
Resistance is measured in Ω
Ohms*

COMPONENTS

CAUSE

RESISTANCE





$$\text{Resistance } (\Omega) = \frac{\text{Potential difference (V)}}{\text{Current (A)}}$$

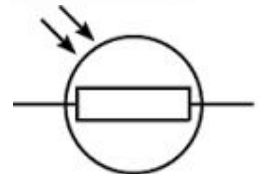
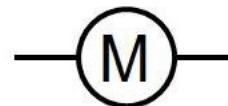
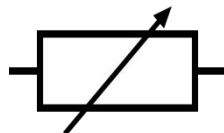
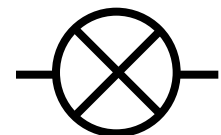
$$V = \frac{E}{C}$$

Ammeters and voltmeters don't cause resistance. They are used to calculate the resistance.

COMPONENTS

CAUSE

RESISTANCE



LOb: Understand and investigate the resistance in a circuit.

Keywords: resistance, Ohm's law, current, potential difference (voltage), electrons

Learning outcomes:

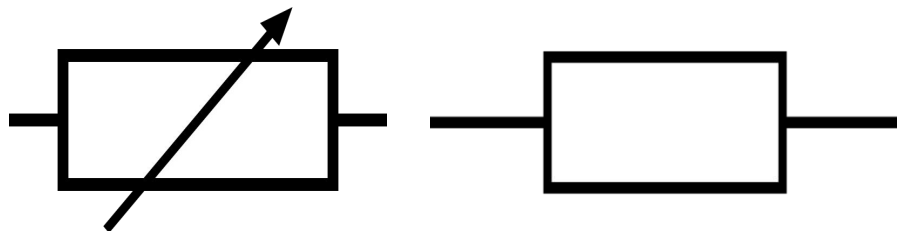
✓ Recall what resistance is and its unit

□ Investigate the resistance in a circuit making note of variables and the safety

Variable Resistor



Resistor



Did you know... resistors come in different forms but all have a basic job. The variable resistor can change its resistance

To calculate the resistance we need to know the current and potential difference using an ammeter and voltmeter.

FACT: As you change the length of wire the resistance changes.

VARIABLES

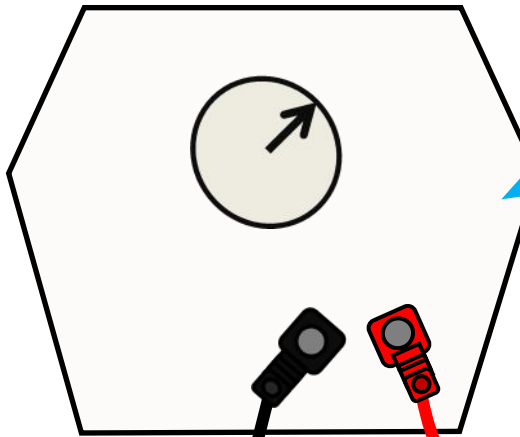
Independent - We will change the length of the wire

Dependent - We will measure the potential difference and current

Control - what we keep the wire, power, time to measure the same



Power Pack



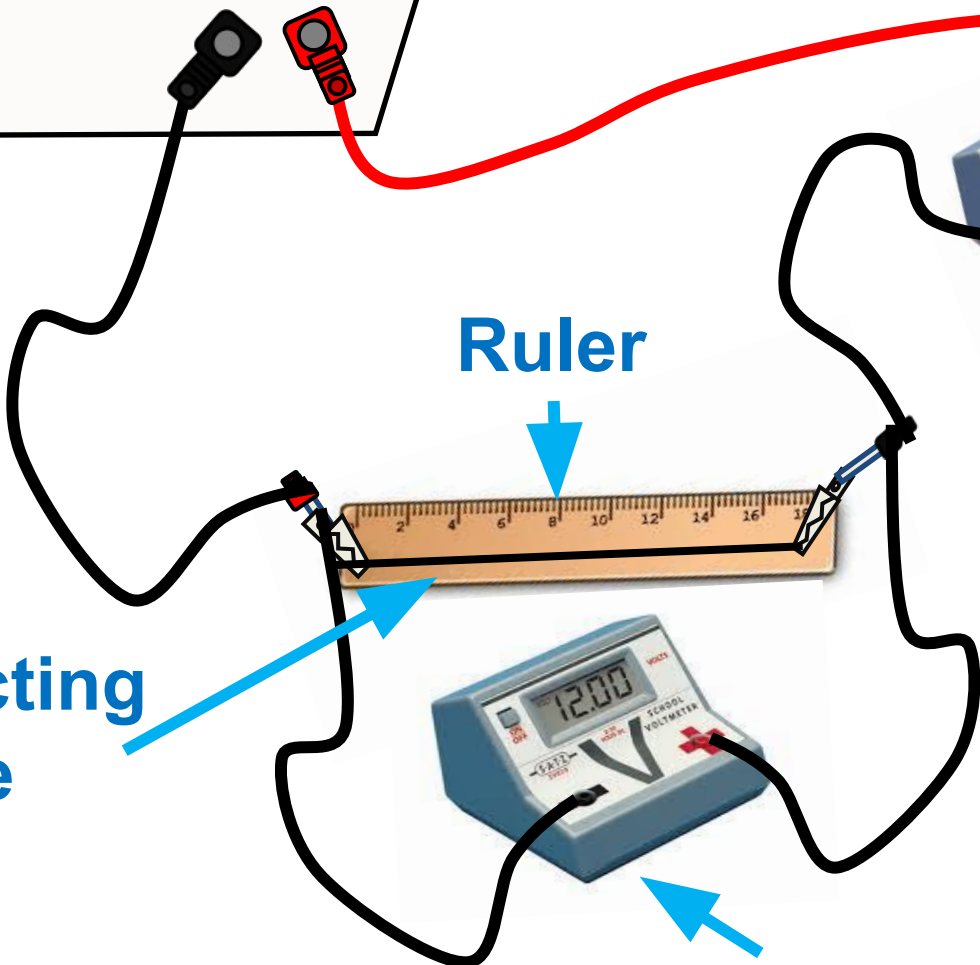
Ruler



Ammeter



Conducting wire



Voltmeter



Aim: Investigate the resistance of different lengths of wire

Prediction: As the length of the wire increases the resistance will...

Equipment:

Dependent (What you measured/recorded)

Independent (What you changed)

Control (What you kept the same)

Method:

- 1.
- 2.
- 3.
- 4.

Safety:

Results:

(units)	(units)

Graph:

Y axis



X axis

Conclusion:

Evaluation:

Safety:

.Don't measure at a low length of wire as the current increases too much which makes the heat increase.

2. Remove clips from wire when not measuring

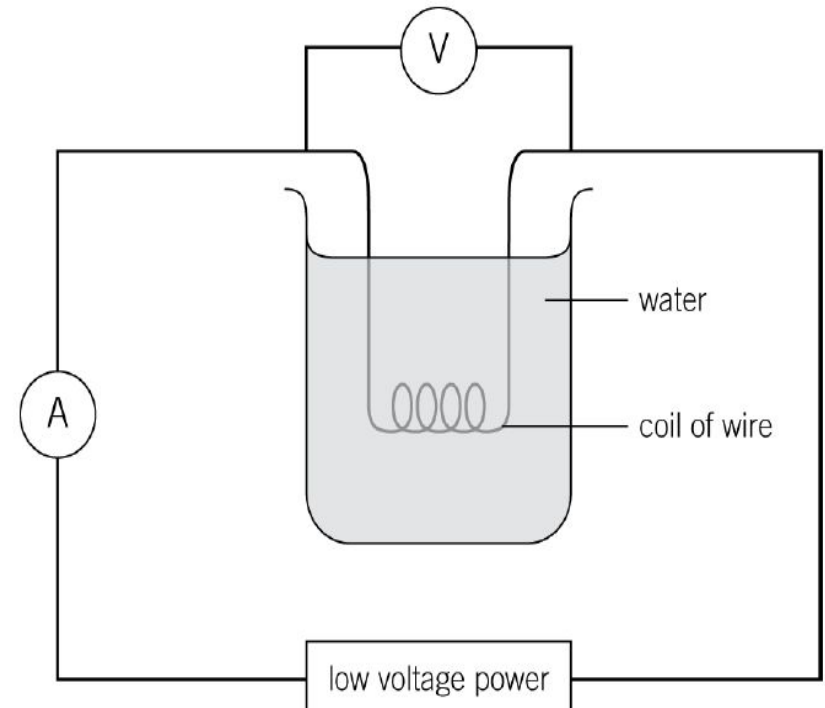
3. Low voltage power



Safety

This was the apparatus used in an experiment to measure the resistance of the coiled wire.

There are two safety elements to this practical what are they?

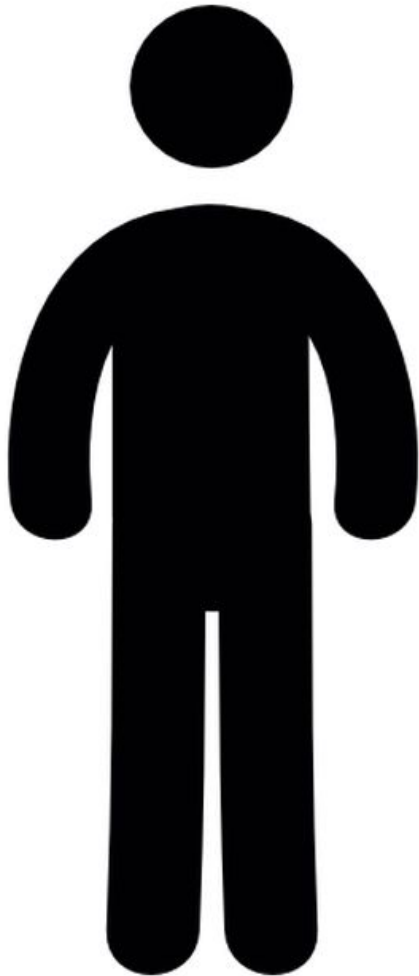


Results Table

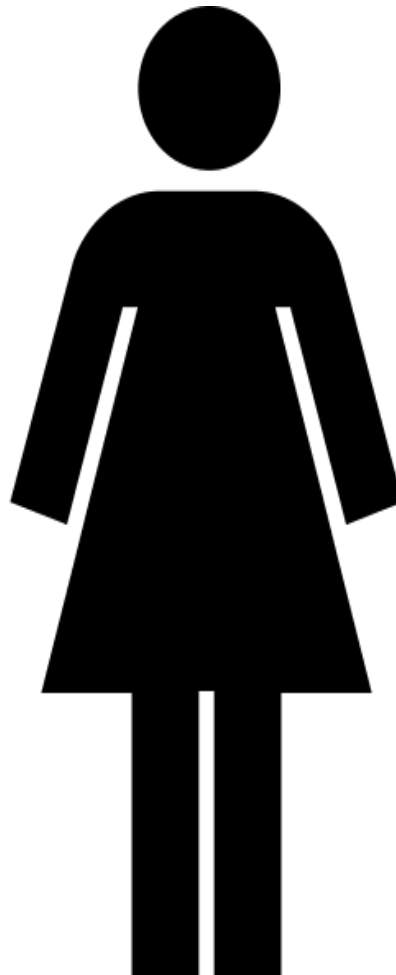
Length of wire	Voltage	Current	Resistance
100			
90			
80			
70			
60			
50			
40			
30			

Responsibilities

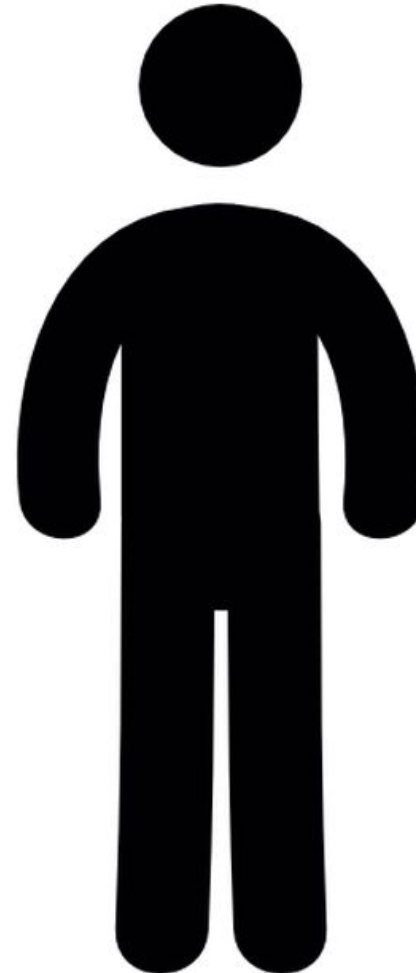
Equipment
Manager



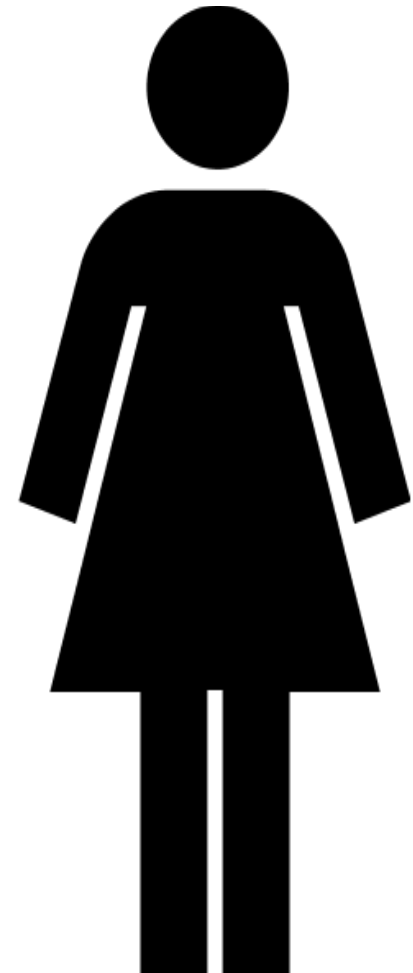
Results
Coordinator



Expectations
Adjudicator



Experiment
Overseer



Aim: Measuring Resistance of different lengths of wire

Prediction: As the length of wire decreases the resistance decreases

Equipment: Nichrome wire, ammeter, voltmeter, battery, metre ruler, croc clips, wires

Dependent (What you measured/recorded)

Voltage
Current

Independent (What you changed)

The length of wire

Control (What you kept the same)

Type of resistance wire
Battery power
Time power is on

Method: make sure all equipment is mentioned in method

1. Set up equipment

2. Measure voltage and current at 100cm

3. Move croc clip to 90cm make sure the wire is taut and measure V and A again

4. Repeat at 80, 70, 60, 50, 40, 30 cm

Results:

(units)	(units)

Graph:

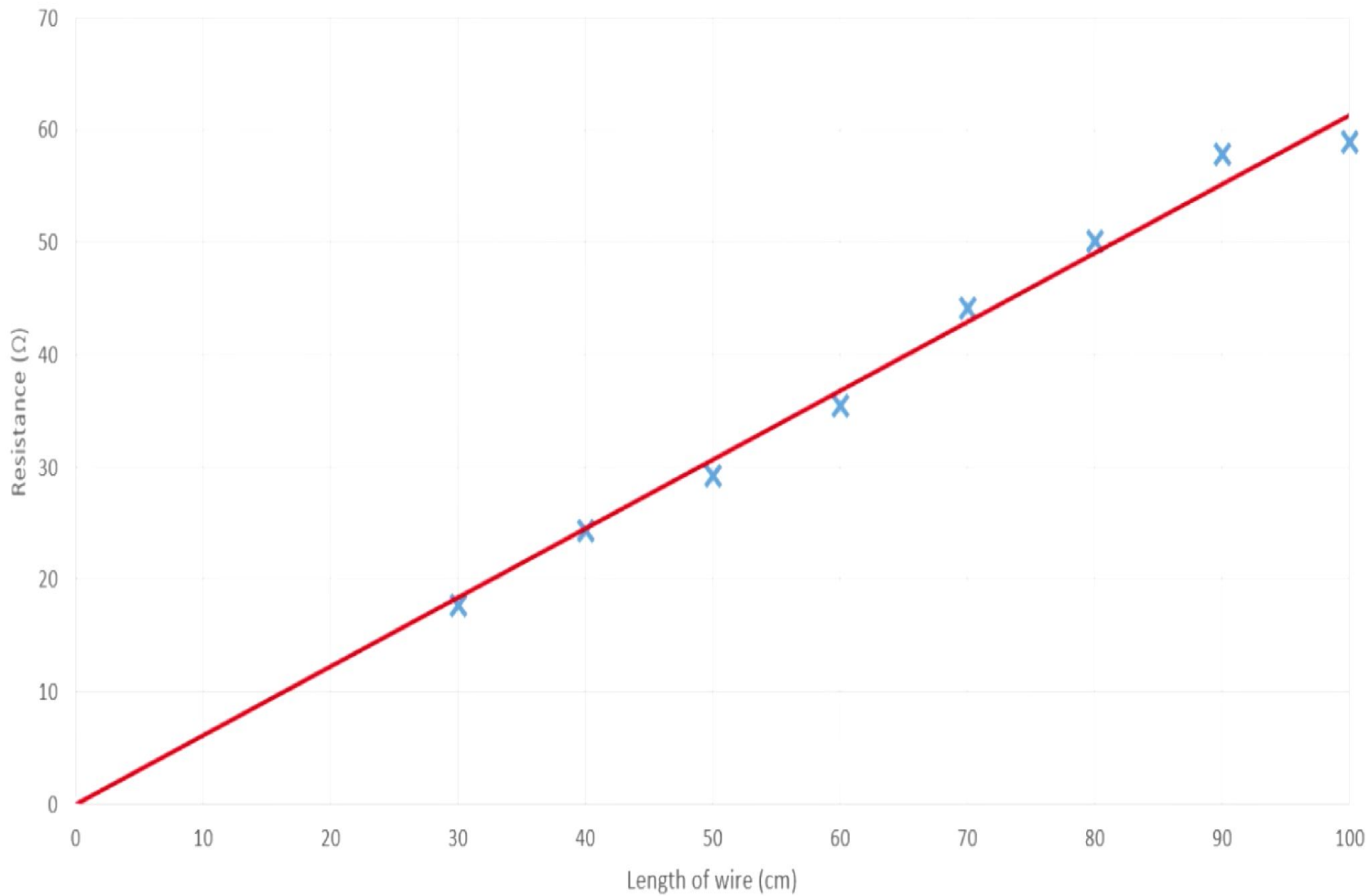
Y axis



Conclusion:

Evaluation:

Resistance vs length of wire



Plenary: Describe what this picture is showing. Use your knowledge of all three

